TEXORA · LECTURE SHEET

Process Control in Comber and Preparatory Process

Comber control is noil control: holding the extracted short-fibre percentage exactly at the planned level, head by head, while keeping lap preparation even enough for the comber to work on. This sheet covers lap weight control, the noil test and its corrections, and sliver quality after combing.

YM-4101 · PROCESS CONTROL IN SPINNING · TOPIC 6 · SPECIALIZATION — YARN MANUFACTURING

1. The preparatory process: lap evenness

The comber grips a fringe of lap; any weight variation across or along the lap becomes uneven gripping, uneven noil and uneven sliver. Preparation control therefore centres on lap weight and its CV (the lap built in YM-3207 Topic 3: 16 doublings of 3.94 g/m sliver at draft 1.05 → 60.0 g/m).

WORKED EXAMPLE — LAP WEIGHT UNIFORMITY TEST

Five 1-metre lap lengths weigh: 59.2, 60.8, 60.1, 59.6, 60.4 g.

Mean = 300.1 / 5 = **60.02 g/m** — on the 60.0 g/m target.

Standard deviation = 0.64 g/m → CV = 0.64/60.02 × 100 = **1.06%**.

Against a working limit of about 1.0–1.2% CV for comber lap, this is acceptable but near the edge: watch the trend. Note the division of duties — the mean is set by doublings and draft at the lap machine; the CV is set by the evenness of the slivers fed and the doubling count, so a CV problem is chased at the cards and drawframes, not at the comber.

2. The noil test

Noil percentage is measured by running a known mass of lap through a head (or collecting the noil over a timed run at known feed) and weighing the extracted noil:

$$\text{Noil (\%)} = \text{noil mass} / \text{lap mass fed} \times 100$$

WORKED EXAMPLE — NOIL OUT OF CONTROL, AND THE CORRECTION

The plan (YM-3207 Topic 8) sets noil at 16%. A control run feeds 240.0 g of lap and collects 43.2 g of noil.

Measured noil = 43.2 / 240.0 × 100 = **18.0%** — two full points high.

At the plan's comber duty that excess is not trivial: 2% of 2427 kg/day of lap ≈ **49 kg/day** of extra fibre leaving as noil, most of it spinnable fibre that should have stayed in the sliver.

Likely causes, in check order: (1) feed per nip has drifted short (4.6 mm instead of the planned 5.0 mm) — each fringe is combed through more cycles per unit length, extracting more; (2) detaching setting closer than standard; (3) top comb penetration too deep; (4) lap lighter than nominal, so the same extraction is a larger percentage.

Correction applied: feed per nip reset to **5.0 mm** and detaching setting verified; retest required before the machine is signed back into control. One head reading differently from its neighbours on the same lap indicates a head-specific mechanical cause; all heads high together indicates a setting or lap cause.

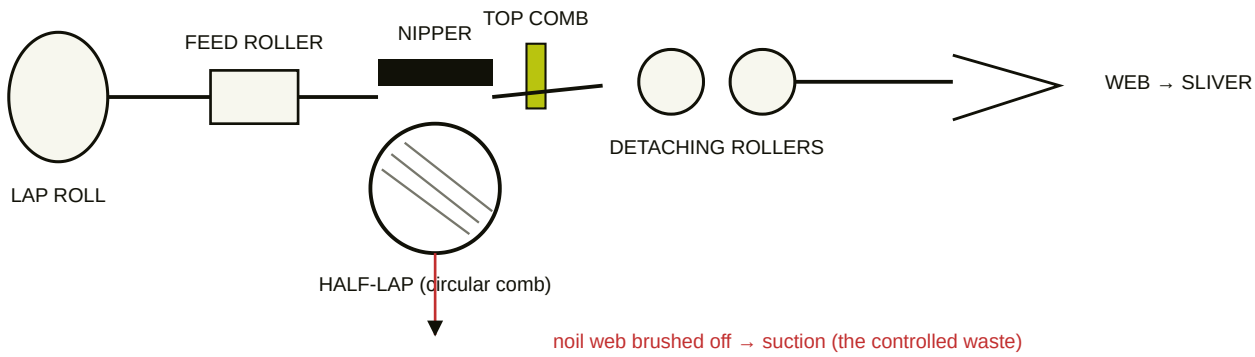


Figure 1 — One combing head in schematic: lap fringe gripped by the nipper, combed by the half-lap, cleaned at its trailing end by the top comb, and detached into the web; short fibres leave below as noil.

3. What moves noil — and what else it moves

ADJUSTMENT	NOIL DIRECTION	SIDE EFFECTS TO RE-CHECK
Feed per nip ↓ (shorter)	Noil ↑	Production per head ↓; sliver weight shifts
Detaching setting closer	Noil ↑	Longer effective combing; watch fibre damage
Top comb penetration deeper	Noil ↑ slightly, neps ↓	Excess causes fibre rupture and sliver faults

Lap weight heavier	Noil % ↓ (same mass extracted from more feed)	Sliver weight and draft balance change

Every noil correction changes production and sliver weight (YM-3207 Topic 3 showed why), so a noil adjustment is always followed by the full verification set: noil retest, sliver weight test, sliver U%.

4. Combed sliver quality

- **Sliver U%** — combing should deliver the most even sliver in the mill, typically U% at or below about 3.0; periodic faults point to damaged detaching rollers or feed irregularity.
- **Mean fibre length in sliver** — verified periodically (AFIS or sorter): the whole economic point of combing; if length has not risen against card sliver, noil is being taken in fibre damage rather than short-fibre removal.
- **Nep level** — should fall sharply versus card sliver; neps surviving combing are usually fibre neps the top comb settings are not catching.
- **Head-to-head uniformity** — noil and sliver weight compared across all 8 heads; combing faults are frequently single-head faults.

Glossary

Preparatory process — sliver-lap and ribbon-lap machines (often with a drawframe) that build the comber lap. **Noil** — short fibres, neps and trash extracted by the comber, quoted as % of lap fed. **Feed per nip** — lap length advanced per combing cycle; a primary noil control. **Detaching setting** — the nipper-to-detaching-roller distance at closest approach. **Top comb** — the stationary comb acting on the trailing end of the fringe during detachment. **Half-lap** — the needle clothing segment of the circular comb. **Head** — one of the (usually eight) identical combing units on a machine.

Key points

• Lap control: mean set at preparation (60.0 g/m in the example), CV 1.06% against a ~1% working limit.
• Noil test: 43.2 g from 240.0 g = 18.0% against a 16% plan — two points of noil ≈ 49 kg/day of extra fibre loss at plan duty.
• Corrections in order: feed per nip, detaching setting, top comb, lap weight — then retest the full set.
• Single head off-line = mechanical cause; all heads together = setting or lap cause.
• Combing must be proved by length gain and nep fall in the sliver, not by noil percentage alone.

Practice

1. A noil run feeds 300 g of lap and collects 39 g of noil. Calculate the noil % and state whether it meets a 14% target.
2. Lap weights per metre are 61.5, 60.9, 62.2, 61.0, 61.8 g. Calculate the mean and comment against a 60.0 g/m nominal. (SD = 0.55 g/m — also give the CV.)
3. Noil is correct at 16% but sliver U% is poor on one head only. Give the most likely causes and the reasoning that excludes a settings cause.

Answer key

1. Noil = 39/300 × 100 = **13.0%** — inside the 14% target (below it; check that short-fibre removal is still adequate via sliver length tests before accepting a lower noil as a saving).
2. Mean = (61.5+60.9+62.2+61.0+61.8)/5 = 307.4/5 = **61.48 g/m** — 2.5% heavy against nominal; the lap machine's doublings or draft need correction, since comber sliver weight will follow. CV = 0.55/61.48 × 100 = **0.89%** — evenness itself is fine; only the level is wrong.
3. Settings are common to the machine's control (and noil, which shares them, is correct), so a one-head fault is mechanical: damaged or eccentric detaching roller on that head, a worn top comb or half-lap segment, or a bent feed roller. The reasoning: a setting cause acts on all heads together and would move noil as well.